

### 3 Ancient Greek Mathematics

#### 3.1 Overview of Ancient Greek Civilization & Early Philosophy

Ancient Greek civilization dates from around 800 BC, centered on the peninsula between the Adriatic Sea (to the west) and the Aegean (to the east) that forms the mainland of the modern Greek state. There was no central controlling empire. Semi-independent city-states were connected by trade and culture, but largely governed themselves. Accomplished seafarers, the Greeks extended their reach round the northern and eastern coasts of the Mediterranean, from Iberia (Spain) to the Black Sea, Anatolia (western Turkey), and Egypt, building and capturing a network of city-states and trading outposts.



Greek Territory around 500 BC

Philip of Macedon ‘unified’ the Greek peninsula just before his death in 336 BC. His son, Alexander the Great, subsequently launched a massive military campaign conquering Persia, Egypt, Babylon, and western India; he died in Babylon in 323 BC. Alexander left various governors to manage his conquered territories: some of these political structures endured for centuries (Egypt’s Ptolemaic dynasty), whereas others were overthrown after only a few years (India/Pakistan). While Alexander’s conquests did not produce a long-lasting centralized Greek empire, they were effective at expanding the reach of Greek culture and philosophy, and brought external ideas into the Greek tradition.

The core part of Greek territory was absorbed by the Roman empire around 146 BC. In line with typical Roman practice, those who agreed to accept Roman governors and taxation could eventually become Roman citizens.<sup>7</sup> As such, the Greek culture of inquiry and scholarship staid largely intact under Roman rule. Greek learning was central to the later Byzantine (eastern Roman) empire centered on Byzantium/Constantinople/Istanbul, which lasted until the capture

<sup>7</sup>While this might sound reasonable, resistance wasn’t realistic, and a significant period of enslavement might first have to be endured...

of Constantinople by the Islamic Ottomans in 1453. For centuries prior, Islamic scholarship had itself been significantly influenced by the ancient Greeks. On consequence of the fall of Constantinople was the exodus of scholars to Rome, helping to fuel the nascent European Renaissance.

## Early Greek Thought

Greek mathematics is part of a much wider philosophy encompassing a change of emphasis from practicality to abstraction. One reason for this was the Greek blending of religion/mysticism with natural philosophy: a desire to describe the natural world while preserving the (seeming) perfection and logic in the gods' design.

Early Greek inquiry into natural phenomena involved the personification of nature (e.g., sky = man, earth = woman). By 600 BC, philosophers were attempting to describe natural phenomena in terms of logical structures. For example, some viewed matter as being comprised of the 'four elements' (fire, earth, water, air) combined in the correct proportions. While the system of the world was considered divinely-designed, explanations relying on the whims of the gods came to be discouraged.

While the Greeks certainly used mathematics for practical purposes, philosophers idealized logic and were unhappy with approximations. This led to the development of *axiomatics*, *theorems* and *proof*, concepts for which there is scant pre-Greek evidence. The ancient Greek language is indeed the source of three words of critical importance:

**Mathematics** *Mathematos* (μαθήματος) meant knowledge or learning. The term covered essentially anything that might be taught in Greek schools. The stem *math* (knowledge) is preserved in other modern usage: e.g. *polymath*.

**Geometry** Literally *earth-measure*, this is a combination of two terms:

*Gi* (γη) Dates from pre-5<sup>th</sup> century BC, meaning *land*, *earth* or *soil*. Capitalized (Γη) it could refer to the *Earth* (as a goddess).

*Metron* (μέτρον) A *weight* or *measure*, a *dimension* (length, width, etc.), or the *metre* (rhythm) in music.

**Theorem** From *theoreo* (θεωρέω), meaning 'I contemplate/consider.' In a mathematical context this become *theoremata* (θεωρήματα): a proposition to be proved.

## Greek Education and Scholarship

Ancient Greece had several schools, mostly private and open only to men. Typically arithmetic was taught until age 14, followed by geometry and astronomy until age 18.

The most famous scholars of ancient Greece were the Athenian trio of Socrates, Plato and Aristotle, each of whom taught his successor,<sup>8</sup> and whose writings became central to both Western and Islamic philosophical tradition. Plato's *Academy* in Athens provided the model for centuries of schooling; the centrality of geometry to Plato's curriculum was evidenced by the famous inscription above its entryway: "Let none ignorant of geometry enter here."

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<sup>8</sup>The birth of Socrates to the death of Aristotle covered 470–322 BC.

## Ancient Greek Enumeration

The Greeks had two primary forms of enumeration.

**Attic Greek** The Attic system (Attica refers to the area around Athens) dates from 7-800 BC. Strokes were used for 1–4, with larger numerals using the first letter of the words for 5, 10, 100, 1000 and 10000. For example,

- Πεντε (*pente*) means five, so Π denoted the number 5.
- Δεκα (*deca*) means ten, so Δ represented 10.
- Η (*hekatón*), Χ (*khilias*) and Μ (*urias*) denoted 100, 1000 and 10000 respectively.
- Larger numbers were written using combinations of these symbols, with larger numerals on the left: e.g., ΧΗΗΙΙΙ = 1207. This is similar to both Egyptian hieroglyphs and Roman numerals. Unlike the latter, Attic Greek did not use pre-subtraction: thus 9 was written ΙΙΙΙΙ instead of ΙΔ.

In modern Greece, Attic numerals are still used occasionally for formal purposes, similarly to how other western countries might use Roman numerals (e.g. construction date of a building).

**Ionic Greek** Ionia refers to the mid Anatolian coast (modern-day Turkey). This system dates from around 400 BC and uses the Ionic Greek alphabet, an approach possibly copied from Egyptian hieratic enumeration. The Ionic numerals slowly displaced the Attic system with most Greek states having adopted them by 1AD.

The Ionic alphabet included several unfamiliar symbols whose shapes and usage changed over time: *digamma* (Ϝ), *stigma* (Ϛ), *qoppa* (ϙ, Ϟ), and *sampi* (Ϙ, ϙ).

Larger numbers were depicted in various ways. A left subscript mark like a comma denoted thousands and/or Μ (with superscripts) for multiples of 10000. For example,

$$35268 = ,\lambda,\epsilon\sigma\zeta\eta = \overset{\gamma}{M},\epsilon\sigma\zeta\eta$$

|   |      |    |      |     |      |
|---|------|----|------|-----|------|
| 1 | α    | 10 | ι    | 100 | ρ    |
| 2 | β    | 20 | κ    | 200 | σ    |
| 3 | γ    | 30 | λ    | 300 | τ    |
| 4 | δ    | 40 | μ    | 400 | υ    |
| 5 | ε    | 50 | ν    | 500 | φ    |
| 6 | Ϝ, Ϛ | 60 | ξ    | 600 | χ    |
| 7 | ζ    | 70 | ο    | 700 | ψ    |
| 8 | η    | 80 | π    | 800 | ω    |
| 9 | θ    | 90 | ϙ, Ϟ | 900 | Ϙ, ϙ |

The Greeks also employed Egyptian fractions, denoting reciprocals with a *keraiā*: e.g.,  $\frac{1}{4} = \delta'$  along with a few special symbols like  $\angle'$  for  $\frac{1}{2}$  (perhaps related to the special Egyptian hieroglyph  $\asymp$  for the same). The use of Egyptian fractions dominated in Europe well into the middle ages.

Both the Attic and Ionic systems were suitable for record-keeping but terrible for calculations! Later Greek mathematicians adapted the Babylonian sexagesimal system with Ionic notation, helping cement its modern use of in astronomy, navigation and time-keeping.

Ionic enumeration is still used in modern Greece, though Hindu–Arabic numerals are now more typical. By the middle-ages, a bar was placed over numerals to distinguish them from words, while modern practice is to insert a *keraiā* similarly to how the ancients denoted fractions: e.g.  $89 = \overline{\pi\theta} = \pi\theta'$ .

**Exercises 3.1.** 1. State the number 1789 in both Attic and Ionic notation.

2. Represent  $\frac{8}{9}$  as a sum of distinct unit fractions (Egyptian style). Express the result in (Ionic) Greek notation.

*(The answer to this problem is non-unique)*

3. For tax purposes, the ancient Greeks would approximate the area of a quadrilateral field by multiplying the averages of the two pairs of opposite sides. In one example, the two pairs of opposite sides were given as

$$a = \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{32} \quad \text{opposite} \quad c = \frac{1}{8} + \frac{1}{16}, \quad \text{and,}$$
$$b = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} \quad \text{opposite} \quad d = 1$$

where the lengths are in fractions of of a *schonion*, a measure of approximately 150 feet. Find the average of  $a$  and  $c$ , the average of  $b$  and  $d$ , and thus the approximate area of the field in square *schonion*. The taxman then rounds up the answer to collect a little more tax!